CIRCUIT SYSTEMS

GET IN TOUCH

ACCESS VERIFICATION

USING XOR/XNOR
LOGIC GATES



TEAM MEMBERS

1. (2620040092) – Konjeti Mokshagna
2. (2620040147)- Mohana Lakshmi Amara
3. (2620040158) –Syed Imran Hussain
4. (2620040170) – Dasam Mohan Sri Sai Koushik
5. (2620040173) – Amogha Sragvi Gurram



INTRODUCTION

This project presents the design and simulation of a Digital Access Card Verification System using XOR/XNOR logic gates. The system compares a binary code from an access card with a pre-stored key. Access is granted only when the codes match and the Enter signal is active.

Input
4-bit Card Code + Stored Key + Enter signal

Process
Bitwise comparison using XNOR gates

Output
Green LED (Granted)/ Red LED (Denied)

WORKING PRINCIPLE

An XNOR gate outputs 1 when both inputs are equal. One XNOR is used for each bit. All XNOR outputs are ANDed to create the Match signal. Match is then ANDed with Enter to produce Granted. A NOT gate generates the Denied signal for the red LED

Problem Statement

Design and simulate a digital access card verification system that:

1. Compares a 4-bit binary code from an access card with a pre-stored 4-bit key
2. Uses XOR or XNOR logic gates for bit-wise comparison
3. Produces clear visual outputs (Granted / Denied) using LEDs
4. Is purely combinational and easy to simulate in Logisim
5. Can be implemented on a breadboard with standard logic ICs

Design / Methodology

4-bit Access Card Verification System

1. Bitwise Match (using XNOR)

$M3 = C3 \odot K3 M_3$

$M2 = C2 \odot K2 M_2$

$M1 = C1 \odot K1 M_1$

$M0 = C0 \odot K0 M_0$

2. Overall Match Signal

$Match = M3 \cdot M2 \cdot M1 \cdot M0$

3. Final Outputs

$Granted = Match \cdot Enter$

$Denied = Granted$

Circuit Components (Logisim)

- 4 Input Pins labeled C3, C2, C1, C0 (Card Code)
- 4 Input Pins labeled K3, K2, K1, K0 (Stored Key)
- 1 Input Pin labeled Enter
- 4 XNOR gates
- AND gates (4-input or cascaded 2-input ANDs)
- 1 additional AND gate (for Enter)
- 1 NOT gate
- 2 LEDs (Green = Granted, Red = Denied)

Circuit Components & Wiring

Components (Logisim)

- 4 Input Pins - C3 C2 C1 C0 (Card)
- 4 Input Pins - K3 K2 K1 K0 (Key) 1 Input Pin - Enter
- 4 XNOR Gates
- AND Gates (Match + Enter)
- 1 NOT Gate
- 2 LEDs (Green & Red)

Wiring Instructions

• Connect each Ci and Ki pair to one XNOR gate.

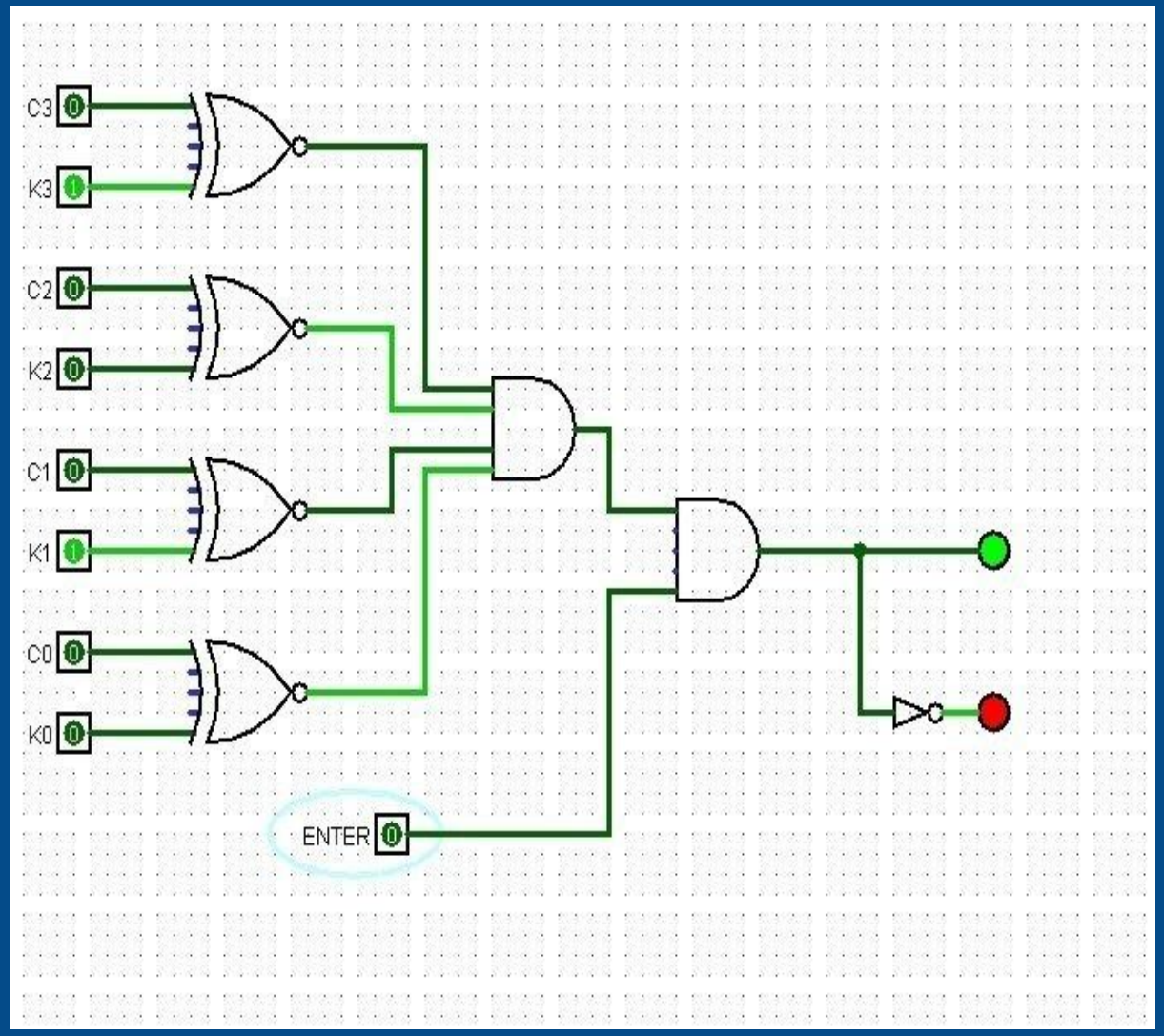
• Feed all four XNOR outputs into the AND gate(s) to produce the Match signal.

• AND the Match signal with the Enter pin to produce the Granted signal.

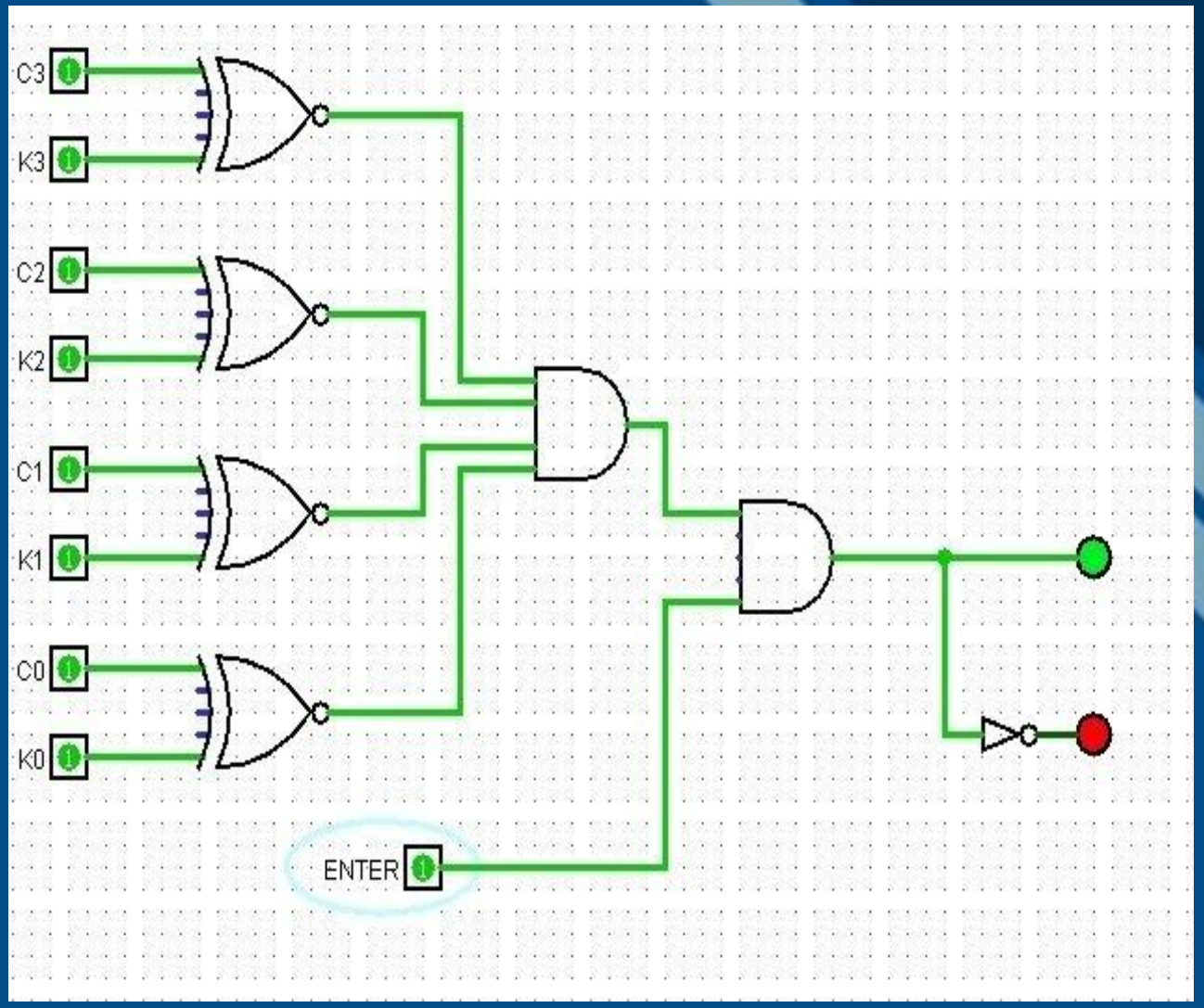
• Connect Granted directly to the Green LED.

• Connect Granted through a NOT gate to the Red LED.

Access Denied



Access accepted



Simulation Results

Card Code	Key Code	Enter	Green LED	Red LED	Result
1010	1010	1	ON	OFF	Access Granted
1010	1010	0	OFF	OFF	No action
1011	1010	1	OFF	ON	Access Denied
0000	1010	1	OFF	ON	Access Denied
1111	1111	1	ON	OFF	Access Granted

How the LEDs Work

Condition	Green LED	Red LED
Correct code + Enter = 1	ON	OFF
Wrong code + Enter = 1	OFF	ON
Enter = 0 (any code)	OFF	OFF

The Green LED (Granted) is driven directly by the final AND gate output. It turns ON only when the Card Code exactly matches the Key Code and the Enter button is pressed. The Red LED (Denied) is driven by the inverted Granted signal. It turns ON when the codes do not match and Enter is pressed. When Enter is not pressed, both LEDs remain OFF.

Conclusion

- The Digital Access Card Verification System successfully demonstrates the use of XOR/XNOR logic for bit-wise comparison.
- The design is simple, fully combinational, easy to simulate in Logisim, and can be implemented with standard logic ICs.
- Simulation in Logisim confirmed correct LED behaviour for all test cases
- The design is simple, fully combinational, and easy to implement in hardware.
- Suitable as an educational project for understanding digital comparators

Thank You

